Results Math (2025) 80:112
© 2025 The Author(s), under exclusive licence to Springer Nature Switzerland AG
1422-6383/25/040001-27
published online May 20, 2025
https://doi.org/10.1007/s00025-025-02431-4

Results in Mathematics



Unimodality Preservation by Ratios of Functional Series and Integral Transforms

Dmitrii Karp, Anna Vishnyakova, and Yi Zhang

Abstract. An elementary, but very useful lemma due to Biernacki and Krzyż (1955) asserts that the ratio of two power series inherits monotonicity from that of the sequence of ratios of their respective coefficients. Over the last two decades it has been realized that, under some additional assumptions, similar claims hold for more general series ratios as well as for unimodality in place of monotonicity. This paper continues this line of research: we consider ratios of general functional series and integral transforms and furnish natural sufficiency conditions for preservation of unimodality by such ratios. Numerous series and integral transforms appearing in applications satisfy our sufficiency conditions, including Dirichlet, factorial and inverse factorial series, Laplace, Mellin and generalized Stieltjes transforms, among many others. Finally, we illustrate our general results by exhibiting certain statements on monotonicity patterns for ratios of some special functions. The key role in our considerations is played by the notion of sign regularity.

Mathematics Subject Classification. Primary 26A48, 44A05; Secondary 15B48, 33C20, 33D15, 33E20.

Keywords. Monotonicity, unimodality, sign-regular kernel, total positivity, quotient of functional series, quotient of integral transforms, hypergeometric ratio, Nuttall Q-function.

1. Introduction

Beginning with the classical lemma due to Biernacki and Krzyż [3] there has been a stream of papers studying the monotonicity pattern of the ratio of two power series

$$\frac{a(x)}{b(x)} = \frac{\sum_{k=0}^{\infty} a_k x^k}{\sum_{k=0}^{\infty} b_k x^k} \tag{1}$$

as the consequence of the corresponding monotonicity pattern of the sequence a_k/b_k. The original result of Biernacki and Krzyż as well as its polynomial version, rediscovered many times, see, for instance, [24, Lemma 1], [16, Theorem 4.4], states that for $b_k > 0$, and assuming positive radius of convergence of both series, the increasing/decreasing character of the sequence a_k/b_k is inherited by the function $a(x)/b(x)$. This result has been extended in three directions - (1) more complicated monotonicity patterns of a_k/b_k; (2) more general functional series, i.e., replacing x^k with some other functional sequence $\phi_k(x)$; (3) replacing series with an integral (Laplace or more general integral transform). The earliest result we are aware of in the first direction is given by Belzunce, Ortega and Ruiz in [2, Lemma 6.4] in connection with application of probability theory in insurance and finance. These authors stated that unimodality (i.e., one change of monotonicity direction) of the sequence a_k/b_k is also preserved by $a(x)/b(x)$. They did not provide a detailed proof but indicated a method based on the notion of total positivity. Their Lemma 6.4 also assumes infinite radii of convergence of both series in (1). This result was thereafter used in several papers sometimes assuming incorrectly that it is valid in the same form for a finite radius of convergence, see details in [37, Introduction]. The situation for the power series was clarified in [37] by Yang, Chu and Wang, who introduced the function $H_{a,b}(x) = b^2(x)(a(x)/b(x))'/b'(x)$ whose sign at the endpoints of the convergence interval is responsible for distinguishing the cases when $a(x)/b(x)$ is monotonic or unimodal. More general ratios of functional series of the form

$$\frac{A(x)}{B(x)} = \frac{\sum_{k=0}^{\infty} a_k \varphi_k(x)}{\sum_{k=0}^{\infty} b_k \varphi_k(x)} \tag{2}$$

have been considered in [25, Lemma 2.2]. Assuming absolute and uniform convergence of both series and their term-wise derivatives on compact subsets of $[0, \infty)$ the authors show that monotonicity of a_k/b_k is inherited by $A(x)/B(x)$ if the sequence of logarithmic derivatives $k \to \varphi_k'(x)/\varphi_k(x)$ is increasing or reversed if $k \to \varphi_k'(x)/\varphi_k(x)$ is decreasing. If fact, they prove more, namely, that the coefficients of the series expansion of $A'(x)B(x) - A(x)B'(x)$ in the functional sequence $\varphi_k(x)$ are all of the same sign. We also note that the same type of monotonicity rule has been independently established for both power series and more general series by Jameson in [18] by using a clever induction argument.

The result for the ratio (2) has been generalized in the recent work [27] to the case when the sequence a_k/b_k is unimodal. The key conditions for preservation of unimodality given in this work are monotonicity of $x \to \varphi_k(x)$ for all k together with monotonicity of $k \to \varphi_k''(x)/\varphi_k'(x)$ for nonnegative x in the convergence domain. They also use the function $H_{A,B}(x) =$

$B^2(x)(A(x)/B(x))'/B'(x)$ whose sign at the endpoint of the convergence interval determines whether the ratio $A(x)/B(x)$ is truly unimodal or merely monotonic.

Finally, in a very recent preprint [28] the authors make one further step and consider the ratio (1), where the sequence a_k/b_k changes monotonicity twice (for example, it increases, then decreases and then again increases). They present conditions in terms of the functions $H_{A,B}$ and $H_{A',B'}$ for the ratio $a(x)/b(x)$ to be monotonic or to have one/two changes in monotonicity.

Note further that similar results for the ratio of the Laplace transforms have been given in [39] and for generic integral transforms in [31] and [27]. Independently, in [13, Lemma 3] the authors established the monotonicity preservation by the ratios of Laplace and generalized Stieltjes transforms. In yet another recent work [29] Mao and Tian give several monotonicity rules for multiple integrals and integrals with variable limits of integration. In [38] the authors consider a more general ratio of the form $(P(x)+A(x))/(P(x)+B(x))$, where $A(x)$ and $B(x)$ are power series and in the recent preprint [26] $A(x)$ and $B(x)$ are integral transforms on time scales. Another approach for studying monotonicity patterns of ratios by using derivatives is based on the so-called L'Hôspital's monotonicity rules [32] and can also be used to give yet one more proof of the Biernacki-Krzyż lemma [11, Theorem 7.3]. A recent overview of the monotonicity rules of this type including integrals with variable limits of integration and generalizations to time scales can be found in [30].

This seemingly exhaustive treatment of the topic is nevertheless incomplete as some important classes of series (for instance, factorial and inverse factorial series and general Dirichlet series) are excluded by conditions in [27] (at least when applied formally). Moreover, these authors require both the numerator coefficients a_k and the denominator coefficients b_k in (2) to be nonnegative, while we retain only the requirements on b_k letting the sign of a_k to be arbitrary. We further remark that the underlying structure behind the conditions in [27] is somewhat obscure. The goal of this paper is to present alternative conditions for unimodality of the ratios of functional series and integral transforms and to reveal that the notion of sign regularity lies at the heart of unimodality preservation property. Furthermore, in some sense our conditions are necessary and sufficient. This is done in the subsequent Section 2. In Section 3 we give a large list of examples of functional sequences and kernels for which our theory holds. In Section 4 we illustrate some of the results by considering the ratios of generalized hypergeometric and basic hypergeometric functions and ratios of Nuttall's Q-functions. Finally, at the end of the paper we included two appendices containing some of the proofs which would distract the reader should they be included in the main text. Our approach bears certain resemblance to the developments in [8], where the authors use total positivity to establish conditions for preservation of quasiconcavity by integration with respect to general measures. In the same paper

and references therein one can find applications in probability and economics. Finally, about half-year after the preliminary version of this paper was made available on arXiv, Zakaria Derbazi published a preprint [10] containing generalization of some of the results of this paper to the case of m-modular sequences and functions. Note, however, that his definition of unimodality is quite different from ours, so that the results can only be compared after the appropriate adaptation.

2. Main results

The key role in our considerations will be played by the notion of sign regularity defined as follows [22, Definition 1.3, Chapter 2, §1].

Definition 2.1. Let X, Y be linearly ordered sets, in this paper - some subsets of $\mathbb{R}$. A function (kernel) $K : X \times Y \to \mathbb{R}$ is called sign regular of order $r \in \mathbb{N}$ ($K \in SR_r$) if there is a sequence of signs $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_r$, each $+1$ or -1 such that

$$\varepsilon_m \begin{vmatrix} K(x_1, y_1) & K(x_1, y_2) & \ldots & K(x_1, y_m) \\ K(x_2, y_1) & K(x_2, y_2) & \ldots & K(x_2, y_m) \\ \vdots & \vdots & \vdots & \ddots \\ K(x_m, y_1) & K(x_m, y_2) & \ldots & K(x_m, y_m) \end{vmatrix} \geq 0 \tag{3}$$

for all $m = 1, \ldots, r$ and any points $x_1 < x_2 < \cdots < x_m$, $y_1 < y_2 < \cdots < y_m$, $x_i \in X$, $y_j \in Y$. If strict inequality holds in the above inequality, the kernel K is called *strictly sign regular* of order r, $K \in SSR_r$. If $\varepsilon_1 = \cdots = \varepsilon_r = 1$, the kernel is *totally positive* of order r, $K \in TP_r$ or *strictly totally positive* ($K \in STP_r$) if all determinants above are strictly positive. We will write $r = \infty$ in any of the above notions, if the corresponding property holds for all natural r.

Remark. Denote by I a convex subset of $\mathbb{R}$ containing more than one point (open or closed interval or semi-interval, open or closed half-axis or the whole axis). Suppose the set X in the above definition is equal to $\mathbb{N}_0$ and the set $Y = I$, so that the kernel $K(x, y)$ represents a sequence of functions $\varphi_n(y) = K(n, y)$, $n \in \mathbb{N}_0$, $y \in I$. Assume further that each function $\varphi_n(y) : I \to \mathbb{R}$ is $(r-1)-$times continuously differentiable, then a sufficient condition for $K(n, y) = \varphi_n(y) \in SSR_r$ is the following: there is sequence of signs $\varepsilon_1, \ldots, \varepsilon_r$ such that for all $m, 1 \leq m \leq r$ and all sequences of indices $0 \leq n_1 < n_2 < \ldots < n_m$ the Wronskians satisfy $\varepsilon_m W(\varphi_{n_1}, \varphi_{n_2}, \ldots, \varphi_{n_m})(x) > 0$ for all $x \in I$. Sequences of functions satisfying these requirements are known as Descartes' sequences or sequences obeying the Descartes' rule of signs [33, Chapter V, Problems 87, 90]. See details below.

Perhaps the most important property of a sign regular kernel is its variation diminishing property. Denote by $S^-(\lambda_n)_{n=0}^{\infty}$ the number of sign changes

of a real sequence $(\lambda_n)_{n=0}^{\infty}$ ignoring zeros; for a continuous function $f(x)$ on an interval I define

$$S^-(f(x))_{x\in I} = \sup S^-(f(x_k))_{k=1}^{n},$$

where supremum is taken over all finite ordered sequences of points in I: $x_1 < x_2 < \cdots < x_n$.

Our main tools are the following theorems, see [22, Chapter V, §3, Theorem 3.1] and [22, Chapter V, §1, Theorem 1.5]

Theorem 2.2. *Suppose for a sequence of real continuous functions $(\varphi_n)_{n=0}^{\infty}$ defined on a convex subset I of $\mathbb{R}$ containing more than one point, the kernel $K(n,x) = \varphi_n(x) \in SR_r$ on $\mathbb{N}_0 \times I$ for some $r \geq 2$. Then for every real sequence $(c_n)_{n=0}^{\infty}$ having no more than $r-1$ sign changes and such that the series*

$$f(x) = \sum_{n=0}^{\infty} c_n \varphi_n(x)$$

converges uniformly on all compact subsets of I, we have

$$S^-(f(x))_{x\in I} \leq S^-(c_n)_{n=0}^{\infty}.$$

Moreover, if $S^-(f(x))_{x\in I} = S^-(c_n)_{n=0}^{\infty} = k \leq r-1$, then the sign patterns of $f(x)$ and $(c_n)_{n=0}^{\infty}$ coincide if $\varepsilon_k\varepsilon_{k+1} = 1$ or are opposite to each other if $\varepsilon_k\varepsilon_{k+1} = -1$. Here $\varepsilon_0 = 1$ and $\varepsilon_m \det(\phi_{n_i}(x_j))_{i,j=1}^{m} \geq 0$ for any finite sequences $n_1 < n_2 < \cdots < n_m$, $x_1 < \cdots < x_m$ with $m \leq r$.

Theorem 2.3. *Suppose the kernel $K(x,y) : I \times J \to \mathbb{R}$ is SR_r for some $r \geq 2$, where I, J are convex subsets of $\mathbb{R}$ containing more than one point. Then for any continuous function $g : J \to \mathbb{R}$ having no more than $r-1$ sign changes and any positive weight $w : J \to (0,\infty)$ such that the integral*

$$f(x) = \int_J K(x,y)g(y)w(y)dy$$

converges uniformly on all compact subsets of I, we have

$$S^-(f(x))_{x\in I} \leq S^-(g(y))_{y\in J}.$$

Moreover, if $S^-(f(x))_{x\in I} = S^-(g(y))_{y\in J} = k \leq r-1$, then the sign patterns of $f(x)$ and $g(y)$ coincide if $\varepsilon_k\varepsilon_{k+1} = 1$ or are opposite to each other if $\varepsilon_k\varepsilon_{k+1} = -1$. Here $\varepsilon_0 = 1$ and $\varepsilon_m \det(K(x_i,y_j))_{i,j=1}^{m} \geq 0$ for any finite sequences $y_1 < y_2 < \cdots < y_m$, $x_1 < \cdots < x_m$ with $m \leq r$.

Remark. The most general form of the variation diminishing property can be found in [22, Chapter V, Theorem 3.1] with both series and the integral above replaced by integration with respect to a general positive measure. The above two theorems are its special cases. However, to avoid unnecessary complications related to the subtleties of the notions of "relevant sign changes" and "nodal zeros", we prefer to present the above two cases separately thus restricting

our attention to sequences and continuous functions which suffices for our purposes.

Remark. The ultimate parts of the above theorems concerning the preservation or reversal of the sign patterns is only stated in [22, Chapter V, Theorem 3.1] for totally positive kernels. However, for the case of matrices the required formulation is contained in [22, Chapter V, §1, Theorem 1.5] (beware of a typo in indexing ε) and can be extended to general kernels repeating *mutatis mutandis* the proof of [22, Chapter V, Theorem 3.1].

Let us fix some terminology. We will say that a continuous function g defined on a convex real interval I containing more than one point is increasing (decreasing) if $g(x_1) \leq g(x_2)$ $(g(x_1) \geq g(x_2))$ for all distinct $x_1 < x_2$, $x_1, x_2 \in I$. This implies that g is either constant or $g(a+) < g(b-)$ $(g(a+) > g(b-))$, where $a = \inf(I)$, $b = \sup(I)$. Increasing or decreasing function will be called monotonic on I. We will further say that g is unimodal if it changes monotonicity no more than once. Hence, there are four possibilities: g is increasing, g is decreasing (or both increasing and decreasing if it is constant), g is first increasing, then decreasing, or g is first decreasing then increasing. If g is unimodal and not monotonic we say that it is strictly unimodal. Similar definition holds for sequences defined on arbitrary subsets of N_0. To cut it short, a function (or a sequence) will be called unimodal if it has no more than one local extremum at interior points of I.

The application of the above theorems to the ratios of functional series and integral transforms is based on the following key observation.

Lemma 2.4. *Let $g : I \to \mathbb{R}$ be a given continuous function. The function g is unimodal if and only if for every $\lambda \in \mathbb{R}$ the function $\mu \to f_\lambda(\mu) = g(\mu) - \lambda$ has no more than 2 sign changes on I, and its sign pattern remains constant for all $\lambda \in \Lambda_2$, where Λ_2 is the subset of $\mathbb{R}$ such that $\mu \to f_\lambda(\mu)$ has exactly 2 sign changes once $\lambda \in \Lambda_2$. In a similar fashion, a real sequence $(d_k)_{k=0}^{\infty}$ is a unimodal if and only if for every $\lambda \in \mathbb{R}$ the sequence $(d_k - \lambda)_{k=0}^{\infty}$ has no more than 2 sign changes and its sign pattern remains constant for all $\lambda \in \Lambda_2$, where Λ_2 is the subset of $\mathbb{R}$ such that $(d_k - \lambda)_{k=0}^{\infty}$ has exactly 2 sign changes once $\lambda \in \Lambda_2$.*

Proof. We will prove the statement about unimodal functions; the case of sequences can be handled analogously.

Suppose that g is a unimodal continuous function. Then g has at most 2 intervals of monotonicity. If g is monotonic on I, then for all $\lambda \in \mathbb{R}$ the function $f_\lambda(\mu) = g(\mu) - \lambda$ has no more than 1 change of sign on I. Suppose now that g is not monotonic, so that it has exactly 2 intervals of monotonicity on I. Without loss of generality, we will assume that there exists $\mu_0 \in I$ such that g is increasing and non-constant on the set $I \cap (-\infty, \mu_0]$, and g is decreasing and non-constant on the set $I \cap [\mu_0, +\infty)$. We denote by $[c, g(\mu_0)] = \overline{g(I \cap (-\infty, \mu_0])}$, and by $[d, g(\mu_0)] = \overline{g(I \cap [\mu_0, +\infty))}$, where

$-\infty \le c < g(\mu_0), \quad -\infty \le d < g(\mu_0)$ (here by $\overline{X}$ we denote the closure of the set X). If $\lambda \le \min(c,d)$, or $\lambda \ge g(\mu_0)$, then the function $\mu \to f_\lambda(\mu)$ has no sign changes on I. If $\lambda \in (\min(c,d), \max(c,d)]$, then the function $\mu \to f_\lambda(\mu)$ has one sign change on I. If $\lambda \in (\max(c,d), g(\mu_0))$, then the function $\mu \to f_\lambda(\mu)$ has two sign changes on I, so that $\Lambda_2 = (\max(c,d)), g(\mu_0))$. For every $\lambda \in \Lambda_2$, the sign pattern of the function $\mu \to f_\lambda(\mu)$ is $(-,+,-)$ and hence remains constant.

Suppose now that for every $\lambda \in \mathbb{R}$ the function $\mu \to f_\lambda(\mu)$ has no more than 2 sign changes on I, and its sign pattern remains constant for all $\lambda \in \Lambda_2$. By the way of contradiction assume that $g(\mu)$ is not unimodal. Then g has at least 3 intervals of monotonicity. Without loss of generality, we can assume that there exist real numbers $\alpha, \beta, \gamma, \delta \in I$ with $\alpha < \beta < \gamma < \delta$ such that

$$g \text{ is increasing on } [\alpha, \beta] \text{ and } g(\alpha) < g(\beta),$$
$$g \text{ is decreasing on } [\beta, \gamma] \text{ and } g(\beta) > g(\gamma),$$
$$g \text{ is increasing on } [\gamma, \delta] \text{ and } g(\gamma) < g(\delta).$$

If $g(\delta) \ge g(\beta)$ then for every $\lambda \in (g(\alpha), g(\beta))$ the function f_λ has at least 3 sign changes on I by Rolle's theorem applied on each subinterval. This contradicts our hypotheses. Thus, $g(\delta) < g(\beta)$. Similar argument yields $g(\alpha) > g(\gamma)$. For $\lambda \in (g(\alpha), g(\beta))$ the function $\mu \to f_\lambda(\mu)$ has 2 sign changes with the pattern $(-,+,-)$ on the interval (α, γ) implying that it has 2 sign changes on the whole I by our assumption and $(\alpha, \gamma) \subset \Lambda_2$. For $\lambda \in (g(\gamma), g(\delta))$ the function f_λ has 2 sign changes on the interval (β, δ) so that it has 2 sign changes on the whole I and $(\beta, \delta) \subset \Lambda_2$. For such values of λ the sign pattern of the function $\mu \to f_\lambda(\mu)$ is $(+,-,+)$. Hence, for different values of $\lambda \in \Lambda_2$ we obtained different sign patterns. This contradicts the hypothesis of a constant sign pattern for all $\lambda \in \Lambda_2$. So, our assumption that g is not unimodal must be wrong. □

Our main results are the following two theorems.

Theorem 2.5. *Suppose for a sequence of real continuous functions $(\varphi_n)_{n=0}^{\infty}$ defined on a convex interval $I \subseteq \mathbb{R}$, the kernel $K(n,x) = \varphi_n(x) \in SR_3$ on $\mathbb{N}_0 \times I$. Suppose both series in the following ratio converge uniformly on compact subsets of I:*

$$F(x) = \frac{\sum_{k=0}^{\infty} a_k \varphi_k(x)}{\sum_{k=0}^{\infty} b_k \varphi_k(x)},$$

where $a_k \in \mathbb{R}$, $b_k > 0$ for all $k \in \mathbb{N}_0$ and the denominator does not vanish for all $x \in I$. If the sequence of quotients $\{a_k/b_k\}_{k=0}^{\infty}$ is a unimodal sequence, then the function F is unimodal. Moreover, if F is strictly unimodal, then it inherits the monotonicity pattern of the sequence $\{a_k/b_k\}_{k=0}^{\infty}$ if $\varepsilon_2\varepsilon_3 > 0$ or reverses it if $\varepsilon_2\varepsilon_3 < 0$, where $\varepsilon_2, \varepsilon_3$ are defined in (3).

A companion theorem for integral transforms takes the form

Theorem 2.6. *Suppose $I, J \subseteq \mathbb{R}$ are convex intervals. Let $K(x,t)$ be a SR_3 kernel on $I \times J \to \mathbb{R}$ and $w : J \to (0,\infty)$ be a positive weight. Suppose both integrals in the following definition converge uniformly on compact subsets of I:*

$$F(x) = \frac{\int_J K(x,t)A(t)w(t)dt}{\int_J K(x,t)B(t)w(t)dt},$$

where $A : J \to \mathbb{R}$ and $B : J \to (0,\infty)$. If the function $t \to A(t)/B(t)$ is unimodal, then the function $x \to F(x)$ is unimodal. Moreover, if F is strictly unimodal, then it inherits the monotonicity pattern of $t \to A(t)/B(t)$ if $\varepsilon_2\varepsilon_3 > 0$ or reverses it if $\varepsilon_2\varepsilon_3 < 0$, where $\varepsilon_2, \varepsilon_3$ are defined in (3).

Remark. All scenarios present in the formulation of these theorems can be realized: $F(x)$ maybe increasing, decreasing or strictly unimodal on I in the direction specified by the theorem. To verify which scenario is realized for a given sequences a_k, b_k, it suffices to verify the sign of the derivative of F at the endpoints of I: $F'(a+)$ and/or $F'(b-)$, where $a = \inf I$, $b = \sup I$.

We will only prove Theorem 2.5, the proof of Theorem 2.6 is almost identical.

Proof. Assume without loss of generality that $\phi_k(x) \geq 0$ on I (otherwise factor out -1). For each $x \in I$ and arbitrary real λ we have

$$F(x) - \lambda = \frac{\sum_{k=0}^{\infty}(a_k - \lambda b_k)\varphi_k(x)}{\sum_{k=0}^{\infty} b_k\varphi_k(x)}. \tag{4}$$

The number of sign changes of $F(x) - \lambda$ on I is equal to the number of sign changes of the function $G(x,\lambda) := \sum_{k=0}^{\infty}(a_k - \lambda b_k)\varphi_k(x)$ on I (as a function of x) due to the condition $b_k > 0$. The function G is a linear combination of a sequence of functions $(\varphi_k(x))_{k=0}^{\infty}$ with coefficients

$$c_k(\lambda) := (a_k - \lambda b_k) = \Big(\frac{a_k}{b_k} - \lambda\Big)b_k.$$

Clearly, $\operatorname{sign}(c_k(\lambda)) = \operatorname{sign}(a_k/b_k - \lambda)$ due to positivity of b_k . By our assumptions, the sequence $(a_k/b_k)_{k=0}^{\infty}$ is a unimodal sequence, so that by Lemma 2.4 $S^-((c_k(\lambda))_{k=0}^{\infty}) \leq 2$. Moreover, for all λ such that the sequence of coefficients $(c_k(\lambda))_{k=0}^{\infty}$ has exactly 2 changes of sign its sign pattern remains constant. Hence, we are in the position to apply Theorem 2.2 yielding that the number of sign changes of $F(x) - \lambda$ on I is less than or equal to $S^-((c_k(\lambda))_{k=0}^{\infty}) \leq 2$. Moreover, by the same theorem for all λ such that the function $F(x) - \lambda$ has exactly 2 sign changes, the sign pattern of the functions $F(x) - \lambda$ coincides with that of the sequences $(c_k(\lambda))_{k=0}^{\infty}$ if $\varepsilon_3\varepsilon_2 = 1$ or is reversed if $\varepsilon_3\varepsilon_2 = -1$. As $\varepsilon_3\varepsilon_2$ does not depend on λ, by Lemma 2.4, we conclude that for every $x \in I$ the function $F(x)$ is a unimodal function of x possessing the monotonicity pattern stated in the theorem. □

3. Examples

1. $K(x,y) = x^y$ is well-known (and easily seen via the Vandermonde determinant) to be STP_∞ on $(0,\infty)\times(-\infty,\infty)$. This recovers the case of ratios of two power series and ratios of the Mellin transforms
$$F(y) = \frac{\int_0^\infty x^{y-1}A(x)dx}{\int_0^\infty x^{y-1}B(x)dx}.$$
If $A(x)/B(x)$ is unimodal, then so is $F(y)$ and the monotonicity pattern of $A(x)/B(x)$ is preserved by $F(y)$ when it is strictly unimodal (= non-monotonic).
2. The kernel $K(x,y) = \exp(xy) \in STP_\infty$ on $(-\infty,\infty)\times(-\infty,\infty)$. Hence, Theorem 2.5 is applicable to ratios of Dirichlet series:
$$F(x) = \frac{\sum_{k=0}^\infty a_k e^{\lambda_k x}}{\sum_{k=0}^\infty b_k e^{\lambda_k x}},$$
where $\lambda_0 < \lambda_1 < \cdots$. If $F(x)$ is strictly unimodal it inherits the monotonicity pattern of $\{a_k/b_k\}$. In a similar fashion, the ratios of two-sided and one-sided Laplace transforms (by restricting the domain of integration to $(-\infty,0)$ and changing variable $x \to -x$)
$$F(y) = \frac{\int_{-\infty}^\infty e^{xy}A(x)dx}{\int_{-\infty}^\infty e^{xy}B(x)dx} \text{ and } G(y) = \frac{\int_0^\infty e^{-xy}A(x)dx}{\int_0^\infty e^{-xy}B(x)dx}$$
satisfy Theorem 2.6, thus recovering some of the results of [39]. Note that the kernel $K(x,y) = e^{-xy}$ has the signature $(+,-,-)$ so that the monotonicity pattern of $A(x)/B(x)$ is preserved by $G(y)$ when it is not monotonic.
3. The kernel $K(x,y) = (x+y)^{-\alpha} \in STP_\infty$ on $(0,\infty)\times(0,\infty)$ for each $\alpha > 0$ [6, (1.6)], so that Theorem 2.5 is applicable to the following ratio of series
$$F(x) = \frac{\sum_{k=0}^\infty a_k (x+k)^{-\alpha}}{\sum_{k=0}^\infty b_k (x+k)^{-\alpha}}$$
and Theorem 2.6 is applicable to the ratio of the generalized Stieltjes transforms
$$F(y) = \int_0^\infty \frac{A(x)dx}{(x+y)^\alpha} \Big/ \int_0^\infty \frac{B(x)dx}{(x+y)^\alpha}.$$
4. The kernel $K(x,y) = \Gamma(x+y) \in STP_\infty$ on $(0,\infty)\times(0,\infty)$, see [15, Theorem 2.11] and Example 10 below. Hence, the kernel $\hat{K}(n,x) = (x)_n = \Gamma(x+n)/\Gamma(x) \in STP_\infty$ on $\mathbb{N}_0\times(0,\infty)$ and Theorem 2.5 is applicable to the ratio of factorial series:
$$F(x) = \frac{\sum_{k=0}^\infty a_k (x)_k}{\sum_{k=0}^\infty b_k (x)_k}.$$

The monotonicity pattern of $\{a_k/b_k\}$ is preserved by $F(x)$ if it is not monotonic. Note that

$$b_0F'(0^+) = \sum_{k=1}^{\infty} b_k(k-1)!\Big(\frac{a_k}{b_k} - \frac{a_0}{b_0}\Big).$$

and $F(x+1)-F(x)$ is eventually negative (positive) if $\{a_k/b_k\}$ is eventually decreasing (increasing) according to Lemma B.1 proved in Appendix B4 to this paper. As $F(x)$ is unimodal if $\{a_k/b_k\}$ is unimodal by Theorem 2.5 it cannot oscillate and $F(x+1)-F(x) < 0$ $(F(x+1)-F(x) > 0)$ for large x implies that $F(x)$ is decreasing (increasing) for large x. Hence, if $\{a_k/b_k\}$ is first increasing and then decreasing, then so is $F(x)$ if $F'(0^+) > 0$ or $F(x)$ is monotonically decreasing if $F'(0^+) < 0$. Similar result holds if $\{a_k/b_k\}$ is first decreasing and then increasing. The case $F'(0^+) = 0$ may require further investigation.
The beta kernel $K(x,y) = B(x,y) = \Gamma(x+y)/[\Gamma(x)\Gamma(y)] \in STP_\infty$ on $(0,\infty)\times(0,\infty)$ by a similar argument, so that one may consider the ratio of the functional series in the sequence $B(x,n)$ on $(0,\infty)\times\mathbb{N}_0$ instead.

5. The argument in [15, Theorem 2.11] for the gamma kernel in the previous example works just as well for the incomplete gamma kernels $K_1(x,y) = \gamma(x+y,\alpha)$ and $K_2(x,y) = \Gamma(x+y,\alpha)$, $\alpha > 0$, where

$$\gamma(z,\alpha) = \int_0^\alpha t^{z-1}e^{-t}dt,\ \Gamma(z,\alpha) = \int_\alpha^\infty t^{z-1}e^{-t}dt$$

are the lower and the upper incomplete gamma functions, respectively. Hence, $K_i(x,y) \in STP_\infty$ on $(0,\infty)\times(0,\infty)$ for $i = 1,2$. This implies that the incomplete Pochhammer symbols [36] are also STP_∞ on $\mathbb{N}_0\times(0,\infty)$:

$$\hat{K}_1(x,n) = (x,\alpha)_n = \frac{\gamma(x+n,\alpha)}{\Gamma(x)} \text{ and } \hat{K}_2(x,n) = [x,\alpha]_n = \frac{\Gamma(x+n,\alpha)}{\Gamma(x)},$$

and Theorem 2.5 is applicable to the ratio of "incomplete factorial series". Some interesting special functions of communication theory like Nuttall and Marcum Q-functions can be defined using such series [4].
6. It follows from [33, Part V, Chap. I, Problem 77] and [22, Chapter 3, Lemma 2.2] that the kernel $\hat{K}(n,x) = 1/(x)_n \in SSR_\infty$ on $\mathbb{N}_0\times(0,\infty)$. We were unable to locate the more general fact in the literature, which we believe (and verified numerically) to be true: the kernel $K(x,y) = 1/\Gamma(x+y) \in SSR_\infty$ on $(0,\infty)\times(0,\infty)$. The property required here is that $\hat{K}(n,x)$ is SSR_3 with the signature $(+,-,-)$ can be proved directly by taking the limit $q\to 1$ in Lemma A.2 proved in Appendix A. Hence, Theorem 2.5 is applicable to the ratio of the inverse factorial series:

$$F(x) = \frac{\sum_{k=0}^\infty a_k/(x)_k}{\sum_{k=0}^\infty b_k/(x)_k}.$$

Monotonicity pattern of $\{a_k/b_k\}$ is preserved by $F(x)$ if it is not monotonic. Note, further, that

$$b_0^2 F'(x) = \frac{b_0 b_1}{x^2}\left(\frac{a_0}{b_0} - \frac{a_1}{b_1}\right) + O(1/x^3) \text{ as } x \to +\infty$$

and, after some calculation (see details in Lemma B.2 in Appendix B):

$$\left(\sum_{k=1}^{\infty} \frac{b_k}{(k-1)!}\right)^2 F'(0+) = \sum_{k=1}^{\infty} \frac{b_0 b_k}{(k-1)!}\left(\frac{a_0}{b_0} - \frac{a_k}{b_k}\right) + \sum_{k=1}^{\infty}\sum_{j=1}^{k-1} \frac{b_k b_j (H_{j-1} - H_{k-1})}{(k-1)!(j-1)!}\left(\frac{a_k}{b_k} - \frac{a_j}{b_j}\right),$$

where H_n denotes the n-th harmonic number. Hence, if a_k/b_k is first increasing and eventually decreasing, then $F(x)$ is decreasing for large x and either monotonically decreasing on $(0,\infty)$ if $F'(0+) < 0$ or first increasing and then decreasing if $F'(0+) > 0$. The case $F'(0+) = 0$ requires further investigation.

7. For $0 < q < 1$ the q-gamma function is defined by [14, (1.10.1)], [19, (21.16)] (see also [17, (1.1)])

$$\Gamma_q(z) = (1-q)^{1-z}\frac{(q;q)_\infty}{(q^z;q)_\infty},\ (a;q)_n = \prod_{j=0}^{n-1}(1-aq^j).$$

We were unable to find in the literature any proof of the following claim: $K(x,y) = \Gamma_q(x+y) \in STP_\infty$. This can probably be established using the basic composition formula [22, Lemma 1.1, Chapter 3, §1] and the representation [19, (21.6)]

$$\Gamma_q(z) = \int_0^\infty x^{z-1} E_q^{-qx} d_q x,$$

similarly to item 9. Here the integral is Jackson's q integral [14, section 1.11], [19, Chapter 19] and E_q^y is the second q-exponential function [19, (9.10)]. The equivalence of this formula to the above definition of q-Gamma can be found in [19, Corollary of Proposition 21.1]. See also [14, Exercise 1.20]. However, for the purposes of this paper we do not need this claim in its full generality, it suffices that $\hat{K}(n,x) = (q^x;q)_n \in STP_3$ on $\mathbb{N}_0 \times (0,\infty)$ which we demonstrate in Lemma A.1 in Appendix A. Hence, Theorem 2.5 is applicable to the ratio of q-factorial series:

$$F(x) = \frac{\sum_{k=0}^{\infty} a_k (q^x;q)_k}{\sum_{k=0}^{\infty} b_k (q^x;q)_k}.$$

If $F(x)$ is strictly unimodal its monotonicity pattern is inherited from that of the sequence $\{a_k/b_k\}$.

8. We further conjecture that $K(x,y) = 1/\Gamma_q(x+y) \in SSR_\infty$, $0 < q < 1$. The claim sufficient for the purposes of this paper, namely that $\hat{K}(n,x) = 1/(q^x;q)_n \in SSR_3$ on $\mathbb{N}_0 \times (0,\infty)$ with the sign pattern $(+,-,-)$ is proved in Lemma A.2 in Appendix A. Hence, Theorem 2.5 is applicable to the ratio of the inverse q-factorial series:

$$F(x) = \frac{\sum_{k=0}^{\infty} a_k/(q^x;q)_k}{\sum_{k=0}^{\infty} b_k/(q^x;q)_k}.$$

If $F(x)$ is strictly unimodal its monotonicity pattern is inherited from that of the sequence $\{a_k/b_k\}$.

9. According to [23, (11)] we have ($\mathbf{c} = (c_1,\dots,c_p)$, $\mathbf{d} = (d_1,\dots,d_p)$)

$$\prod_{i=1}^{p} \frac{\Gamma(x+c_i)}{\Gamma(x+d_i)} = \int_0^1 t^x \left\{ G^{p,0}_{p,p}\left(t \,\middle|\, \begin{matrix}\mathbf{d}\\ \mathbf{c}\end{matrix}\right) + \delta_1 I_{\{\mu=0\}} \right\} \frac{dt}{t},$$

where $\mu = \sum_{i=1}^{p}(d_i - c_i) \geq 0$, δ_1 denotes the unit mass concentrated at 1 and $I_{\{\mu=0\}} = 1$ if $\mu = 0$ and $I_{\{\mu=0\}} = 0$ otherwise. Here $G^{p,0}_{p,p}$ stands for a particular case of Meijer's G function which we prefer to call the Meijer-Nørlund function, see [23, p.139] for a definition. For current purposes we only need one property of this function [23, section 2]:

$$\text{if } v(t) = \sum_{j=1}^{p} (t^{c_j} - t^{d_j}) \geq 0 \text{ for } t \in (0,1), \text{ then } G^{p,0}_{p,p}\left(t \,\middle|\, \begin{matrix}\mathbf{d}\\ \mathbf{c}\end{matrix}\right) \geq 0 \text{ for } t \in (0,1). \tag{5}$$

Furthermore, convenient sufficient conditions for $v(t) \geq 0$ are the following:

$$\begin{aligned} &0 \leq c_1 \leq c_2 \leq \cdots \leq c_p,\ 0 \leq d_1 \leq d_2 \leq \cdots \leq d_p, \\ &\text{and } \sum_{i=1}^{k} c_i \leq \sum_{i=1}^{k} d_i \text{ for } k = 1,2\dots,p. \end{aligned} \tag{6}$$

These and other conditions for non-negativity of $G^{p,0}_{p,p}$ above can be found in [23, section 2]. Assuming that G function is the integrand is non-negative we can write

$$K(x,y) = \prod_{i=1}^{p} \frac{\Gamma(x+y+c_i)}{\Gamma(x+y+d_i)} = \int_0^1 K_1(x,t) K_2(t,y) d\sigma(t),$$

where $K_1(x,t) = t^x$, $K_2(t,y) = t^y$ and $d\sigma(t)$ is a non-negative measure supported on $[0,1]$. Hence, by the basic composition formula [22, Lemma 1.1, Chapter 3, §1], the kernel $K(x,y)$ is STP_∞ on $(0,\infty)\times(0,\infty)$,

so that the kernel

$$K(n,x)=\prod_{i=1}^{p}\frac{(x+c_i)_n}{(x+d_i)_n}\in STP_\infty \text{ on } \mathbb{N}_0\times(0,\infty).$$

Hence, Theorem 2.5 is applicable to the ratio

$$F(x)=\frac{\sum\limits_{k=0}^{\infty}a_k\prod\limits_{i=1}^{p}[(c_i+x)_k/(d_i+x)_k]}{\sum\limits_{k=0}^{\infty}b_k\prod\limits_{i=1}^{p}[(c_i+x)_k/(d_i+x)_k]},$$

and if $F(x)$ is not monotonic it inherits monotonicity pattern of $\{a_k/b_k\}$.

10. We can extend the previous example as follows. Suppose $p\geq 0$ and $q\geq 1$. Consider

$$K(x,y)=\prod_{i=1}^{p}\frac{\Gamma(x+y+c_i)}{\Gamma(x+y+d_i)}\prod_{j=1}^{q}\Gamma(h_j+x+y).$$

By the previous example if $\sum_{i=1}^{p}(d_i-c_i)>0$, then

$$\prod_{i=1}^{p}\frac{\Gamma(x+y+c_i)}{\Gamma(x+y+d_i)}=\mathcal{M}(\rho(t))(x+y)$$

where $\mathcal{M}$ is the Mellin transform and

$$\rho(t)=I_{[0,1]}(t)G_{p,p}^{p,0}\left(t\,\middle|\,\begin{matrix}\mathbf{d}\\ \mathbf{c}\end{matrix}\right).$$

Hence,

$$K(x,y)=\mathcal{M}(\rho(t)*\gamma_1(t)*\cdots*\gamma_q(t))(x+y),\ \gamma_j(t)=t^{h_j}e^{-t},$$

where $*$ is the Mellin convolution:

$$h*g(x)=\int_0^\infty h(t)g\Big(\frac{x}{t}\Big)\frac{dt}{t}.$$

Hence, if $v(t)=\sum_{j=1}^{p}(t^{c_j}-t^{d_j})\geq 0$ on $(0,1)$ and $h_j\geq 0$, $K(x,y)=\mathcal{M}(f)(x+y)$, where the function $f(t)=\rho(t)*\gamma_1(t)*\cdots*\gamma_q(t)\geq 0$ on $(0,\infty)$, so that $K(x,y)\in STP_\infty$ by the basic composition formula as in the previous example. In particular, the kernel

$$K(x,y)=\prod_{i=1}^{q}\Gamma(x+y+h_i)\in STP_\infty$$

for any $h_i\geq 0$ and any natural q and so is $K(n,x)=\prod_{i=1}^{q}(h_i+x)_n$ on $\mathbb{N}_0\times(0,\infty)$.

11. Numerical experiments suggest that the kernel

$$K(x,y)=\frac{\Gamma(c+x+y)}{\Gamma(d+x+y)}$$

is also SR_∞ when $c > d$. In particular, it is SR_3 with the signature $(+,-,-)$. We were unable to locate any results for sign regularity of this kernel in the literature. The above kernel is an example of the product of two sign regular Hankel kernels (i.e. the kernels of the form $K(x,y) = F(x+y)$) and we believe that such product is sign regular in general. The corresponding result for TP_r matrices is known [12, Theorem 4.5], but $F(x_i + y_j)$ is not, in general, a Hankel matrix. Apoorva Khare informed us that he has a method for approximating Hankel kernels by Hankel matrices which may lead to a proof of the required property. We postpone the details to another publication.

12. Suppose $\mathbf{a} = (a_1, \dots, a_p)$ with $a_i > 0$, $i = 1, \dots, p$, and $\mathbf{m} = (m_1, m_2, \dots, m_p)$ comprises non-negative integers. Then, according to [34, Theorem 3.2], the kernel

$$K_1(x,y) = {}_pF_p\left(\begin{matrix}\mathbf{a}\\ \mathbf{a}+\mathbf{m}\end{matrix}\middle| xy\right) \in STP_\infty \text{ on } \mathbb{R}^2.$$

Note also that for any positive $\mathbf{a}$, $\mathbf{b}$ the kernel

$$K_2(x,y) = {}_pF_q\left(\begin{matrix}\mathbf{a}\\ \mathbf{b}\end{matrix}\middle| xy\right) \in STP_\infty \text{ on } (0,\infty)\times(0,\infty)$$

due to [22, p.101, below (1.9) in Chapter 3]. This implies that Theorem 2.6 is applicable to the ratios of the integral transforms

$$F(x) = \frac{\int_{-\infty}^{\infty} K_1(x,t)A(t)w(t)dt}{\int_{-\infty}^{\infty} K_1(x,t)B(t)w(t)dt} \text{ and } G(x) = \frac{\int_{0}^{\infty} K_2(x,t)A(t)w(t)dt}{\int_{0}^{\infty} K_2(x,t)B(t)w(t)dt}$$

and the monotonicity pattern of $A(x)/B(x)$ is preserved both by $F(x)$ and $G(x)$ when they are not monotonic.

13. Consider the kernel $K(n,x) = I_n(x)$, where I_ν is the modified Bessel function [29, p.217], [5, p.101]. According to [5, Theorem 1.5] this kernel is STP_∞ on $\mathbb{N}_0 \times (0,\infty)$. Hence, Theorem 2.5 is applicable to the ratio of the modified Bessel function expansions of the form

$$F(x) = \frac{\sum_{k=0}^{\infty} a_k I_k(x)}{\sum_{k=0}^{\infty} b_k I_k(x)}.$$

Similarly, restricting the interval I in Theorem 2.6 to non-negative integers, we conclude that

$$G(k) = \frac{\int_{0}^{\infty} I_k(t)A(t)w(t)dt}{\int_{0}^{\infty} I_k(x)B(t)w(t)dt}$$

is unimodal on $\mathbb{N}_0$ if so is $A(x)/B(x)$. If $F(x)$ $(G(k))$ is not monotonic it inherits the monotonicity pattern of $\{a_k/b_k\}$ $(A(x)/B(x))$.

14. Let $L_j^\alpha(t)$ be the generalized Laguerre polynomials [9, p.3] with $\alpha > -1$ and $j = 0, 1, \dots$. According to [9, Theorem 2] the matrix $[L_j^\alpha(t_i)]_{0\le i,j\le n}$ is STP for each $n \in \mathbb{N}$ and $\alpha > -1$ if the points t_i satisfy $t_n < t_{n-1} < \cdots < t_1 < t_0 < 0$. This implies that ordering the points ascending $\hat{t}_i = t_{n-i}$

the kernel $K(j,t) = L_j^{\alpha}(t)$ is SSR_∞ with the sign pattern $(-1)^{r(r-1)/2}$ (where r is the size of the corresponding minor), in particular, it is SSR_3 with the sign pattern $(+,-,-)$. Hence, Theorem 2.5 is applicable to the ratio of the expansions in the Laguerre polynomials when the argument is outside of the interval of orthogonality:

$$F(x) = \frac{\sum_{k=0}^{\infty} a_k L_k^{\alpha}(x)}{\sum_{k=0}^{\infty} b_k L_k^{\alpha}(x)}$$

where $\alpha > -1$, $x < 0$, and if $F(x)$ is not monotonic it inherits the monotonicity pattern of $\{a_k/b_k\}$.

4. Applications

Given an m-tuple $\mathbf{c} = (c_1, \ldots, c_m)$, we will use the shorthand notation $(\mathbf{c}+\mu)_k = \prod_{i=1}^{m}(c_i+\mu)_k$ and the product is understood to be 1 if $m = 0$. Define

$$F(\mu) := \frac{{}_pF_q\left(\begin{matrix}\mathbf{c}+\mu, \mathbf{a}_1 \\ \mathbf{d}+\mu, \mathbf{b}_1\end{matrix}\middle| x\right)}{{}_sF_t\left(\begin{matrix}\mathbf{c}+\mu, \mathbf{b}_2 \\ \mathbf{d}+\mu, \mathbf{a}_2\end{matrix}\middle| x\right)} = \frac{\sum\limits_{k=0}^{\infty} f_k(\mathbf{c}+\mu)_k/(\mathbf{d}+\mu)_k}{\sum\limits_{k=0}^{\infty} g_k(\mathbf{c}+\mu)_k/(\mathbf{d}+\mu)_k}, \tag{7}$$

where $\mathbf{c}$ or $\mathbf{d}$ can be missing. By writing $\mathbf{a} = (\mathbf{a}_1, \mathbf{a}_2)$, $\mathbf{b} = (\mathbf{b}_1, \mathbf{b}_2)$ we will have

$$\frac{f_k}{g_k} = \frac{(\mathbf{a}_1)_k(\mathbf{a}_2)_k}{(\mathbf{b}_1)_k(\mathbf{b}_2)_k} = \frac{(\mathbf{a})_k}{(\mathbf{b})_k}.$$

The following fact is well-known and is straightforward to verify: both log-concavity and log-convexity of a sequence imply its unimodality (and if a log-concave sequence is not monotone, then it first increases and then decreases while for log-convex sequence it is reverse).

Log-concavity of $\{f_k/g_k\}$ reduces to the inequality (similarly for log-convexity)

$$\frac{(\mathbf{a}+k-1)}{(\mathbf{b}+k-1)} \geq \frac{(\mathbf{a}+k)}{(\mathbf{b}+k)} \text{ for } k = 1, 2, \ldots$$

A sufficient condition for the above inequality to hold is the decrease of the rational function

$$R_{m,n}(x) = \frac{(\mathbf{a}+x)}{(\mathbf{b}+x)} = \frac{\prod_{k=1}^{m}(a_k+x)}{\prod_{k=1}^{n}(b_k+x)}$$

on $(0,\infty)$. Here m, n denote the number of components in $\mathbf{a}$ and $\mathbf{b}$, respectively. Let $e_j(\mathbf{a}) = e_j(a_1, \ldots, a_m)$ be the j-th elementary symmetric polynomial. We have a slight extension of Biernaki–Krzyż lemma as follows [20, Lemmas 3,4]:

Lemma 4.1. *If $m \le n$ and*

$$\frac{e_n(\mathbf{b})}{e_m(\mathbf{a})} \le \frac{e_{n-1}(\mathbf{b})}{e_{m-1}(\mathbf{a})} \le \cdots \le \frac{e_{n-m+1}(\mathbf{b})}{e_1(\mathbf{a})} \le e_{n-m}(\mathbf{b}),$$

then the function $R_{m,n}(x)$ is monotone decreasing on $(0,\infty)$. These inequalities hold, in particular, if $0 < a_1 \le a_2 \le \cdots \le a_m$, $0 < b_1 \le b_2 \le \cdots \le b_m$ and $\sum_{j=1}^{k} a_j \le \sum_{j=1}^{k} b_j$ for $k = 1, \ldots, m$.

Remark. Clearly, we can apply the above lemma to $1/R_{m,n}(x)$ and get conditions for $R_{m,n}$ to be increasing and hence for $\{f_k/g_k\}$ to be log-convex (and still unimodal).

Theorem 4.2. *Suppose the kernel*

$$K(\mu, n) = \frac{(\mathbf{c}+\mu)_n}{(\mathbf{d}+\mu)_n} \in SR_3 \text{ on } (0,\infty) \times \mathbb{N}_0$$

and $\mathbf{a} = (\mathbf{a}_1, \mathbf{a}_2)$, $\mathbf{b} = (\mathbf{b}_1, \mathbf{b}_2)$ satisfy the conditions of Lemma 4.1. Then the function $F(\mu)$ defined in (7) is unimodal on $(0,\infty)$. In particular, this is true if the kernel $K(\mu, n)$ is one of those given in Examples 4, 6, 9 and 10.

In some particular cases of the above theorem we can say more. Take $\mathbf{a} = (\mathbf{a}_1, \mathbf{a}_2)$, $\mathbf{b} = (\mathbf{b}_1, \mathbf{b}_2)$ satisfying Lemma 4.1 and consider

$$F_1(\mu) = {}_pF_q\left(\begin{matrix}\mu, \mathbf{a}_1 \\ \mathbf{b}_1\end{matrix}\middle| x\right) \Big/ {}_sF_t\left(\begin{matrix}\mu, \mathbf{b}_2 \\ \mathbf{a}_2\end{matrix}\middle| x\right),$$

so that we are in the framework of Example 4. We conclude that $F_1(\mu)$ is decreasing for large μ by Lemma B.1 while the sign of $F'(0)$ coincides, by an easy calculation, with that of

$$\frac{(\mathbf{a})_1}{(\mathbf{b})_1} {}_{p+1}F_{q+1}\left(\begin{matrix}1, \mathbf{a}_1+1 \\ 2, \mathbf{b}_1+1\end{matrix}\middle| x\right) - \frac{(\mathbf{b}_2)_1}{(\mathbf{a}_2)_1} {}_{s+1}F_{t+1}\left(\begin{matrix}1, \mathbf{b}_2+1 \\ 2, \mathbf{a}_2+1\end{matrix}\middle| x\right).$$

Hence, if this quantity is negative, then $F_1(\mu)$ is monotone decreasing on $(0,\infty)$. If it is positive, $F_1(\mu)$ is first increasing and then decreasing. On the other hand, taking

$$F_2(\mu) = {}_pF_q\left(\begin{matrix}\mathbf{a}_1 \\ \mu, \mathbf{b}_1\end{matrix}\middle| x\right) \Big/ {}_sF_t\left(\begin{matrix}\mathbf{b}_2 \\ \mu, \mathbf{a}_2\end{matrix}\middle| x\right)$$

we get a particular case of Example 6. Thus we conclude that $F_2(\mu)$ is eventually decreasing by the asymptotic formula given in Example 6 (still assuming that $\mathbf{a}$, $\mathbf{b}$ satisfy Lemma 4.1). The derivative $F_2'(0+)$ can be computed by Lemma B.2.

In a similar fashion, by employing Examples 7 and 8 we get conditions for unimodality of the q-hypergeometric ratios

$$\mu \to {}_p\varphi_q\left(\begin{matrix}q^{\mu}, q^{\mathbf{a}_1} \\ q^{\mathbf{b}_1}\end{matrix}\middle| q; x\right) \Big/ {}_s\varphi_t\left(\begin{matrix}q^{\mu}, q^{\mathbf{b}_2} \\ q^{\mathbf{a}_2}\end{matrix}\middle| q; x\right)$$

$$\mu \to {}_p\varphi_q\left(\begin{matrix}q^{\mathbf{a}_1} \\ q^{\mu}, q^{\mathbf{b}_1}\end{matrix}\middle| q; x\right) \Big/ {}_s\varphi_t\left(\begin{matrix}q^{\mathbf{b}_2} \\ q^{\mu}, q^{\mathbf{a}_2}\end{matrix}\middle| q; x\right).$$

Note, however, that for the q case the m-tuples $\mathbf{a}$ and $\mathbf{b}$ in Lemma 4.1 should replaced by the m-tuples:

$$\hat{\mathbf{a}} = (q^{-a_1}-1, q^{-a_2}-1, \dots, q^{-a_m}-1), \quad \hat{\mathbf{b}} = (q^{-b_1}-1, q^{-b_2}-1, \dots, q^{-b_m}-1)$$

as explained in [21, Theorem 1].

As another example consider the Nuttall Q-function [4] which generalizes Marcum Q-function and plays a role in communication theory. It is defined by

$$Q_{\mu,\nu}(a,b) = \int_b^\infty x^\mu e^{-(x^2+a^2)/2} I_\nu(ax)dx \tag{8}$$

with $a > 0$, $b \geq 0$, $\nu > -1$, $\mu > 0$ and I_ν standing for the modified Bessel function of the first kind. Fix $\nu_1 > \nu_2$ and $a_1 \leq a_2$. Using the standard power series expansion of I_ν [29, p.217] we have

$$\frac{I_{\nu_1}(a_1x)}{I_{\nu_2}(a_2x)} = 2^{\nu_2-\nu_2} a_1^{\nu_1} a_2^{-\nu_2} x^{\nu_1-\nu_2} \frac{\sum_{k=0}^\infty a_1^{2k}x^{2k}/[4^k k!\Gamma(k+\nu_1+1)]}{\sum_{k=0}^\infty a_2^{2k}x^{2k}/[4^k k!\Gamma(k+\nu_2+1)]}.$$

First we will show that this ratio is a unimodal function of x on $(0,\infty)$ if $\nu_1 - \nu_2 = 2\ell$, where $\ell \in \mathbb{N}$ (if $\ell = 0$ this ratio is immediately seen to be decreasing). Indeed, in this case the power series in the numerator becomes

$$\sum\nolimits_{n=\ell}^\infty a_1^{2(n-\ell)} x^{2n}/[4^{n-\ell}(n-\ell)!\Gamma(n-\ell+\nu_1+1)]$$

and the sequence of ratios of the power series coefficients at x^{2k} for $k = 0, 1, \dots$ takes the form

$$\underbrace{0,\dots,0}_{\ell \text{ times}}, \frac{4^\ell \ell!\Gamma(\ell+\nu_2+1)}{a_2^{2\ell}\Gamma(2\ell+\nu_2+1)}, \dots, 4^\ell a^{-2\ell}\Big(\frac{a_1}{a_2}\Big)^{2k} \frac{\Gamma(k+1)\Gamma(k+\nu_2+1)}{\Gamma(k-\ell+1)\Gamma(k+\ell+\nu_2+1)}, \dots$$

It follows from the $p = 2$ case of the integral representation in Example 9 (see also [23, Introduction]) that

$$k \to \frac{\Gamma(k+1)\Gamma(k+\nu_2+1)}{\Gamma(k-\ell+1)\Gamma(k+\ell+\nu_2+1)}$$

is decreasing for $k \geq \ell$, so that the ratios of the coefficients at x^{2k}, $k = 0, 1, \dots$ form a sequence which first increases and then decreases. By Theorem 2.5 or [37, Corollary 2.3] we conclude that the same is true for the ratio $I_{\nu_1}(a_1x)/I_{\nu_2}(a_2x)$. Hence, considering the definition (8) as the (truncated) Mellin transform in the variable μ we get the following statement:

Theorem 4.3. *Suppose $b \geq 0$, $\nu_1 - \nu_2 = 2\ell \in \mathbb{N}$ and $0 < a_1 \leq a_2$. Then the ratio*

$$\mu \to \frac{Q_{\mu,\nu_1}(a_1,b)}{Q_{\mu,\nu_2}(a_2,b)}$$

is unimodal on $(0,\infty)$.

In view of the reduction formula [4, p.39]

$$Q_{\mu,\nu}(a,0) = 2^{(\mu-\nu-1)/2} a^{\nu} e^{-a^2/2} \frac{\Gamma((\mu+\nu+1)/2)}{\Gamma(\nu+1)} {}_1F_1\left(\begin{matrix} (\mu+\nu+1)/2 \\ \nu+1 \end{matrix} \middle| \frac{a^2}{2} \right),$$

we obtain the corresponding statement for the ratio of the Kummer functions. Furthermore, based on numerical evidence, we believe that the above ratio of the modified Bessel functions is unimodal not only if $\nu_1 - \nu_2 = 2\ell$ but for all $\nu_1 \geq \nu_2 > -1$.

Acknowledgements

We thank Apoorva Khare for sharing some of his insights regarding sign regularity which was very fruitful and helped to improve the paper. We are further indebted to anonymous referee for spotting numerous inaccuracies in the first draft of this paper and various insightful suggestions. The work of D. Karp was supported by the Bulgarian Ministry of Education and Science, Scientific Programme "Enhancing the Research Capacity in Mathematical Sciences (PIKOM)", No. DO1-67/05.05.2022 and The Simons Foundation International (grant No. 992227, IMI-BAS). The work of Y. Zhang was supported by the NSFC grants No. 12101506 and No. 12371520.

Author contributions All authors wrote the main manuscript text and contributed equally to the paper.

Data Availability Statement No datasets were generated or analysed during the current study.

Declarations

Conflicts of Interest The authors declare no Conflict of interest.

A Sign regularity of some q-Pochhammer kernels

Assume that $k \in \mathbb{N}$, $0 < q < 1$, and x, y are indeterminates. Let $(x;q)_k$ be the q-Pochhammer symbol. We will need the following easily verifiable identity

$$\frac{(x;q)_m - (y;q)_m}{x-y} = -\sum_{j=0}^{m-1} q^j (x;q)_j (yq^{j+1};q)_{m-1-j}, \tag{A.1}$$

where $m \in \mathbb{N}_0$. The proof of the following lemma is inspired by a calculation in [7, Theorem 3.2].

Lemma A.1. *The kernel $K(\mu, n) = (q^{\mu};q)_n \in STP_3$ on $(0,\infty) \times \mathbb{N}_0$.*

Proof. As $K(\mu,n)>0$ we need to verify the signs of the second and the third order determinants. Verification for the second order determinants is rather straightforward and will be omitted. Suppose $\mu_3>\mu_2>\mu_1>0$ are reals and $s>j>i\geq 1$ are positive integers. It remains to prove that

$$\begin{vmatrix} (q^{\mu_1};q)_i & (q^{\mu_2};q)_i & (q^{\mu_3};q)_i \\ (q^{\mu_1};q)_j & (q^{\mu_2};q)_j & (q^{\mu_3};q)_j \\ (q^{\mu_1};q)_s & (q^{\mu_2};q)_s & (q^{\mu_3};q)_s \end{vmatrix} > 0.$$

Using rather standard notation for column and row manipulations (say $C_2-C_1\to C_2$ means that the second column should be replaced by the difference of the second and the first column which does not alter the value of the determinant), we compute

$$\begin{vmatrix} (q^{\mu_1};q)_i & (q^{\mu_2};q)_i & (q^{\mu_3};q)_i \\ (q^{\mu_1};q)_j & (q^{\mu_2};q)_j & (q^{\mu_3};q)_j \\ (q^{\mu_1};q)_s & (q^{\mu_2};q)_s & (q^{\mu_3};q)_s \end{vmatrix} = \begin{bmatrix} C_2-C_1\to C_2 \\ C_3-C_1\to C_3 \end{bmatrix}$$

$$= \begin{vmatrix} (q^{\mu_1};q)_i & (q^{\mu_2};q)_i-(q^{\mu_1};q)_i & (q^{\mu_3};q)_i-(q^{\mu_1};q)_i \\ (q^{\mu_1};q)_j & (q^{\mu_2};q)_j-(q^{\mu_1};q)_j & (q^{\mu_3};q)_j-(q^{\mu_1};q)_j \\ (q^{\mu_1};q)_s & (q^{\mu_2};q)_s-(q^{\mu_1};q)_s & (q^{\mu_3};q)_s-(q^{\mu_1};q)_s \end{vmatrix}$$

$$= \left[R_3-(q^{\mu_1+j};q)_{s-j}R_2\to R_3\right] =$$

$$\begin{vmatrix} (q^{\mu_1};q)_i & (q^{\mu_2};q)_i-(q^{\mu_1};q)_i & (q^{\mu_3};q)_i-(q^{\mu_1};q)_i \\ (q^{\mu_1};q)_j & (q^{\mu_2};q)_j-(q^{\mu_1};q)_j & (q^{\mu_3};q)_j-(q^{\mu_1};q)_j \\ 0 & (q^{\mu_2};q)_s-(q^{\mu_1+j};q)_{s-j}(q^{\mu_2};q)_j & (q^{\mu_3};q)_s-(q^{\mu_1+j};q)_{s-j}(q^{\mu_3};q)_j \end{vmatrix}$$

$$= \left[R_2-(q^{\mu_1+i};q)_{j-i}R_1\to R_2\right]$$

$$= \begin{vmatrix} (q^{\mu_1};q)_i & (q^{\mu_2};q)_i-(q^{\mu_1};q)_i & (q^{\mu_3};q)_i-(q^{\mu_1};q)_i \\ 0 & (q^{\mu_2};q)_j-(q^{\mu_1+i};q)_{j-i}(q^{\mu_2};q)_i & (q^{\mu_3};q)_j-(q^{\mu_1+i};q)_{j-i}(q^{\mu_3};q)_i \\ 0 & (q^{\mu_2};q)_s-(q^{\mu_1+j};q)_{s-j}(q^{\mu_2};q)_j & (q^{\mu_3};q)_s-(q^{\mu_1+j};q)_{s-j}(q^{\mu_3};q)_j \end{vmatrix}$$

$$= (q^{\mu_1};q)_i \begin{vmatrix} (q^{\mu_2};q)_i & \left[(q^{\mu_2+i};q)_{j-i}-(q^{\mu_1+i};q)_{j-i}\right] & (q^{\mu_3};q)_i & \left[(q^{\mu_3+i};q)_{j-i}-(q^{\mu_1+i};q)_{j-i}\right] \\ (q^{\mu_2};q)_j & \left[(q^{\mu_2+j};q)_{s-j}-(q^{\mu_1+j};q)_{s-j}\right] & (q^{\mu_3};q)_j & \left[(q^{\mu_3+j};q)_{s-j}-(q^{\mu_1+j};q)_{s-j}\right] \end{vmatrix}$$

$$= \text{by (A.1)} = (q^{\mu_1};q)_i(q^{\mu_2}-q^{\mu_1})(q^{\mu_3}-q^{\mu_1})q^iq^j \times$$

$$\begin{vmatrix} (q^{\mu_2};q)_i \sum\limits_{k=0}^{j-i-1} q^k(q^{\mu_2+i};q)_k(q^{\mu_1+i+k+1};q)_{j-i-1-k} & (q^{\mu_3};q)_i \sum\limits_{k=0}^{j-i-1} q^k(q^{\mu_3+i};q)_k(q^{\mu_1+i+k+1};q)_{j-i-1-k} \\ (q^{\mu_2};q)_j \sum\limits_{\ell=0}^{s-j-1} q^\ell(q^{\mu_2+j};q)_\ell(q^{\mu_1+j+\ell+1};q)_{s-j-1-\ell} & (q^{\mu_3};q)_j \sum\limits_{\ell=0}^{s-j-1} q^\ell(q^{\mu_3+j};q)_\ell(q^{\mu_1+j+\ell+1};q)_{s-j-1-\ell} \end{vmatrix}$$

$$= (q^{\mu_1};q)_i(q^{\mu_2}-q^{\mu_1})(q^{\mu_3}-q^{\mu_1})q^iq^j \times$$

$$\sum_{k=0}^{j-i-1}\sum_{\ell=0}^{s-j-1} q^{k+\ell}(q^{\mu_1+i+k+1};q)_{j-i-1-k}(q^{\mu_1+j+\ell+1};q)_{s-j-1-\ell} \begin{vmatrix} (q^{\mu_2};q)_{i+k} & (q^{\mu_3};q)_{i+k} \\ (q^{\mu_2};q)_{j+\ell} & (q^{\mu_3};q)_{j+\ell} \end{vmatrix}.$$

As, clearly, $i+k\leq j-1<j+\ell$ each determinant in the summand is positive. □

Lemma A.2. *The kernel* $K(\mu,n)=1/(q^\mu;q)_n\in SSR_3$ *on* $(0,\infty)\times\mathbb{N}_0$ *with the sign pattern* $(+,-,-)$.

Proof. As $K(\mu,n)>0$ we need to verify the signs of the second and the third order determinants. Verification for the second order determinants is rather straightforward and will be omitted. Suppose $\mu_3>\mu_2>\mu_1>0$ are reals and

$s > j > i \geq 1$ are positive integers. It remains to prove that

$$\begin{vmatrix} (q^{\mu_1};q)_i^{-1} & (q^{\mu_2};q)_i^{-1} & (q^{\mu_3};q)_i^{-1} \\ (q^{\mu_1};q)_j^{-1} & (q^{\mu_2};q)_j^{-1} & (q^{\mu_3};q)_j^{-1} \\ (q^{\mu_1};q)_s^{-1} & (q^{\mu_2};q)_s^{-1} & (q^{\mu_3};q)_s^{-1} \end{vmatrix} < 0. \tag{A.2}$$

We compute

$$\begin{aligned}
&\begin{vmatrix} (q^{\mu_1};q)_i^{-1} & (q^{\mu_2};q)_i^{-1} & (q^{\mu_3};q)_i^{-1} \\ (q^{\mu_1};q)_j^{-1} & (q^{\mu_2};q)_j^{-1} & (q^{\mu_3};q)_j^{-1} \\ (q^{\mu_1};q)_s^{-1} & (q^{\mu_2};q)_s^{-1} & (q^{\mu_3};q)_s^{-1} \end{vmatrix} = \begin{bmatrix} C_2 - C_1 \to C_2 \\ C_3 - C_1 \to C_3 \end{bmatrix} \\
&= \begin{vmatrix} (q^{\mu_1};q)_i^{-1} & (q^{\mu_2};q)_i^{-1} - (q^{\mu_1};q)_i^{-1} & (q^{\mu_3};q)_i^{-1} - (q^{\mu_1};q)_i^{-1} \\ (q^{\mu_1};q)_j^{-1} & (q^{\mu_2};q)_j^{-1} - (q^{\mu_1};q)_j^{-1} & (q^{\mu_3};q)_j^{-1} - (q^{\mu_1};q)_j^{-1} \\ (q^{\mu_1};q)_s^{-1} & (q^{\mu_2};q)_s^{-1} - (q^{\mu_1};q)_s^{-1} & (q^{\mu_3};q)_s^{-1} - (q^{\mu_1};q)_s^{-1} \end{vmatrix} \\
&= \left[R_3 - (q^{\mu_1+j};q)_{s-j}^{-1} R_2 \to R_3 \right] \\
&= \begin{vmatrix} (q^{\mu_1};q)_i^{-1} & (q^{\mu_2};q)_i^{-1} - (q^{\mu_1};q)_i^{-1} & (q^{\mu_3};q)_i^{-1} - (q^{\mu_1};q)_i^{-1} \\ (q^{\mu_1};q)_j^{-1} & (q^{\mu_2};q)_j^{-1} - (q^{\mu_1};q)_j^{-1} & (q^{\mu_3};q)_j^{-1} - (q^{\mu_1};q)_j^{-1} \\ 0 & (q^{\mu_2};q)_s^{-1} - (q^{\mu_1+j};q)_{s-j}^{-1}(q^{\mu_2};q)_j^{-1} & (q^{\mu_3};q)_s^{-1} - (q^{\mu_1+j};q)_{s-j}^{-1}(q^{\mu_3};q)_j^{-1} \end{vmatrix} \\
&= \left[R_2 - (q^{\mu_1+i};q)_{j-i}^{-1} R_1 \to R_2 \right] \\
&= \begin{vmatrix} (q^{\mu_1};q)_i^{-1} & (q^{\mu_2};q)_i^{-1} - (q^{\mu_1};q)_i^{-1} & (q^{\mu_3};q)_i^{-1} - (q^{\mu_1};q)_i^{-1} \\ 0 & (q^{\mu_2};q)_j^{-1} - (q^{\mu_1+i};q)_{j-i}^{-1}(q^{\mu_2};q)_i^{-1} & (q^{\mu_3};q)_j^{-1} - (q^{\mu_1+i};q)_{j-i}^{-1}(q^{\mu_3};q)_i^{-1} \\ 0 & (q^{\mu_2};q)_s^{-1} - (q^{\mu_1+j};q)_{s-j}^{-1}(q^{\mu_2};q)_j^{-1} & (q^{\mu_3};q)_s^{-1} - (q^{\mu_1+j};q)_{s-j}^{-1}(q^{\mu_3};q)_j^{-1} \end{vmatrix} \\
&= (q^{\mu_1};q)_i^{-1} \times \begin{vmatrix} a & b \\ c & d \end{vmatrix},
\end{aligned} \tag{A.3}$$

where

$$\begin{aligned}
a &= (q^{\mu_2};q)_i^{-1}(q^{\mu_2+i};q)_{j-i}^{-1}(q^{\mu_1+i};q)_{j-i}^{-1}\left[(q^{\mu_2+i};q)_{j-i} - (q^{\mu_1+i};q)_{j-i}\right], \\
b &= (q^{\mu_3};q)_i^{-1}(q^{\mu_3+i};q)_{j-i}^{-1}(q^{\mu_1+i};q)_{j-i}^{-1}\left[(q^{\mu_3+i};q)_{j-i} - (q^{\mu_1+i};q)_{j-i}\right], \\
c &= (q^{\mu_2};q)_j^{-1}(q^{\mu_2+j};q)_{s-j}^{-1}(q^{\mu_1+j};q)_{s-j}^{-1}\left[(q^{\mu_2+j};q)_{s-j} - (q^{\mu_1+j};q)_{s-j}\right], \\
d &= (q^{\mu_3};q)_j^{-1}(q^{\mu_3+j};q)_{s-j}^{-1}(q^{\mu_1+j};q)_{s-j}^{-1}\left[(q^{\mu_3+j};q)_{s-j} - (q^{\mu_1+j};q)_{s-j}\right].
\end{aligned}$$

By (A.1) , we see that (A.3) is equal to

$$(q^{\mu_1};q)_i^{-1}(q^{\mu_2} - q^{\mu_1})(q^{\mu_3} - q^{\mu_1})q^i q^j \times \begin{vmatrix} u & v \\ f & g \end{vmatrix}, \tag{A.4}$$

where

$$u=(q^{\mu_2};q)_i^{-1}(q^{\mu_2+i};q)_{j-i}^{-1}(q^{\mu_1+i};q)_{j-i}^{-1}\sum_{k=0}^{j-i-1}q^k(q^{\mu_2+i};q)_k(q^{\mu_1+i+k+1};q)_{j-i-1-k},$$

$$v=(q^{\mu_3};q)_i^{-1}(q^{\mu_3+i};q)_{j-i}^{-1}(q^{\mu_1+i};q)_{j-i}^{-1}\sum_{k=0}^{j-i-1}q^k(q^{\mu_3+i};q)_k(q^{\mu_1+i+k+1};q)_{j-i-1-k},$$

$$f=(q^{\mu_2};q)_j^{-1}(q^{\mu_2+j};q)_{s-j}^{-1}(q^{\mu_1+j};q)_{s-j}^{-1}\sum_{\ell=0}^{s-j-1}q^\ell(q^{\mu_2+j};q)_\ell(q^{\mu_1+j+\ell+1};q)_{s-j-1-\ell},$$

$$g=(q^{\mu_3};q)_j^{-1}(q^{\mu_3+j};q)_{s-j}^{-1}(q^{\mu_1+j};q)_{s-j}^{-1}\sum_{\ell=0}^{s-j-1}q^\ell(q^{\mu_3+j};q)_\ell(q^{\mu_1+j+\ell+1};q)_{s-j-1-\ell}.$$

In view of $(q^{\mu_2};q)_i^{-1}(q^{\mu_2+i};q)_{j-i}^{-1}=(q^{\mu_2};q)_j^{-1}$ and $(q^{\mu_3};q)_i^{-1}(q^{\mu_3+i};q)_{j-i}^{-1}=(q^{\mu_3};q)_j^{-1}$, we conclude that that (A.4) is equal to

$$(q^{\mu_1};q)_i^{-1}(q^{\mu_2}-q^{\mu_1})(q^{\mu_3}-q^{\mu_1})q^{i+j}(q^{\mu_1+i};q)_{j-i}^{-1}(q^{\mu_1+j};q)_{s-j}^{-1}(q^{\mu_2};q)_j^{-1}(q^{\mu_3};q)_j^{-1}$$
$$\times\sum_{k=0}^{j-i-1}\sum_{\ell=0}^{s-j-1}q^{k+\ell}(q^{\mu_1+i+k+1};q)_{j-i-1-k}(q^{\mu_1+j+\ell+1};q)_{s-j-1-\ell}\Delta_{k\ell},$$

where

$$\begin{aligned}\Delta_{k\ell}&=\begin{vmatrix}(q^{\mu_2+i};q)_k & (q^{\mu_3+i};q)_k\\ (q^{\mu_2+j};q)_{s-j}^{-1}(q^{\mu_2+j};q)_\ell & (q^{\mu_3+j};q)_{s-j}^{-1}(q^{\mu_3+j};q)_\ell\end{vmatrix}\\&=(q^{\mu_3+j};q)_{s-j}^{-1}(q^{\mu_2+j};q)_{s-j}^{-1}(q^{\mu_3+j};q)_\ell(q^{\mu_2+j};q)_\ell\\&\times\left[(q^{\mu_2+j+\ell};q)_{s-j-\ell}(q^{\mu_2+i};q)_k-(q^{\mu_3+j+\ell};q)_{s-j-\ell}(q^{\mu_3+i};q)_k\right]<0\end{aligned}$$

in view of $\mu_2<\mu_3$. This proves (A.2). □

B Derivative asymptotics for factorial and inverse factorial series

Lemma B.1. *Suppose $b_k>0$ for all $k\geq 0$ and*

$$\frac{a_0}{b_0}\leq\frac{a_1}{b_1}\leq\cdots\leq\frac{a_m}{b_m},\quad\frac{a_m}{b_m}\geq\frac{a_{m+1}}{b_{m+1}}\geq\frac{a_{m+2}}{b_{m+2}}\cdots$$

and at least one inequality in each chain is strict. Define

$$F(x)=\frac{\sum_{k=0}^\infty a_k(x)_k}{\sum_{k=0}^\infty b_k(x)_k}=\frac{A(x)}{B(x)},$$

and assume that each series converges uniformly on all compact subsets of $\mathbb{R}$. Then $F(x+1)-F(x)<0$ for all sufficiently large x.

Proof. We have

$$F(x+1)-F(x)=\frac{A(x+1)B(x)-A(x)B(x+1)}{B(x)B(x+1)}.$$

In view of $(x+1)_k = (x)_k(x+k)/x$, we obtain

$$\begin{aligned}
A(x+1)B(x) - A(x)B(x+1) &= \frac{1}{x}\sum_{k=0}^{\infty} a_k (x)_k (x+k) \sum_{n=0}^{\infty} b_n (x)_n \\
&\quad - \frac{1}{x}\sum_{k=0}^{\infty} a_k (x)_k \sum_{n=0}^{\infty} b_n (x)_n (x+n) \\
&= \frac{1}{x}\sum_{k,n=0}^{\infty} a_k b_n (x)_k (x)_n (k-n) = -\frac{1}{x}\sum_{k,n=0}^{\infty} a_n b_k (x)_k (x)_n (k-n) \\
&= \frac{1}{2x}\sum_{k,n=0}^{\infty} b_k b_n (x)_k (x)_n (k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) \\
&= \frac{S(x)}{2x},
\end{aligned}$$

so that

$$F(x+1) - F(x) = \frac{S(x)}{2xB(x)B(x+1)}$$

and the sign of the left hand side coincides with that of $S(x)$. Decomposing yields:

$$\begin{aligned}
S(x) &= \sum_{k,n=0}^{m-1} b_k b_n (x)_k (x)_n (k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) \\
&+ \sum_{n=0}^{m-1}\sum_{k=m}^{\infty} b_k b_n (x)_k (x)_n (k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) \\
&+ \sum_{k=0}^{m-1}\sum_{n=m}^{\infty} b_k b_n (x)_k (x)_n (k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) + \sum_{k=m}^{\infty}\sum_{n=m}^{\infty} b_k b_n (x)_k (x)_n (k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) \\
&= \sum_{k,n=0}^{m-1} b_k b_n (x)_k (x)_n (k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) + 2\sum_{k=m}^{\infty} b_k (x)_k \sum_{n=0}^{m-1} b_n (x)_n (k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) \\
&+ \sum_{k=m}^{\infty} b_k (x)_k \sum_{n=m}^{\infty} b_n (x)_n (k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) = \sum_{k,n=0}^{m-1} b_k b_n (x)_k (x)_n (k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) \\
&+ \sum_{k=m}^{\infty} b_k (x)_k \left[\sum_{n=0}^{m-1} 2b_n (x)_n (k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) - \sum_{n=m}^{\infty} b_n (x)_n (n-k)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right)\right].
\end{aligned}$$

From the last expression we get

$$\begin{aligned}
\frac{S(x)}{[(x)_m]^2} &= \sum_{k,n=0}^{m-1} b_k b_n \frac{(x)_k (x)_n}{(x)_m (x)_m}(k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) \\
&+ \sum_{k=m}^{\infty} b_k \frac{(x)_k}{(x)_m}\left[\sum_{n=0}^{m-1} 2b_n \frac{(x)_n}{(x)_m}(k-n)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right) - \sum_{n=m}^{\infty} b_n \frac{(x)_n}{(x)_m}(n-k)\left(\frac{a_k}{b_k} - \frac{a_n}{b_n}\right)\right].
\end{aligned}$$

Clearly, we can make the first sum as small as we wish by making x sufficiently large. Next, the expression in brackets in the second sum is eventually negative (for large x) , since the first sum tends to zero while the second sum is positive for all $x > 0$ and is monotonically increasing, because for some (at least one) $n > m$ the coefficient at $(x)_n/(x)_m$ is strictly positive, so that the second sum in brackets tends to $+\infty$ as x increases. This implies that $S(x)$ is negative for sufficiently large x which establishes our claim. □

Lemma B.2. *Define*

$$F(x) = \frac{\sum_{k=0}^{\infty} a_k/(x)_k}{\sum_{k=0}^{\infty} b_k/(x)_k}.$$

and assume that each series converges uniformly on all compact subsets of $(0, \infty)$. *Then*

$$\left(\sum_{k=1}^{\infty} \frac{b_k}{(k-1)!}\right)^2 F'(0+) = \sum_{k=1}^{\infty} \frac{b_0 b_k}{(k-1)!}\left(\frac{a_0}{b_0} - \frac{a_k}{b_k}\right) + \sum_{k=1}^{\infty}\sum_{j=1}^{k-1} \frac{b_k b_j (H_{j-1} - H_{k-1})}{(k-1)!(j-1)!}\left(\frac{a_k}{b_k} - \frac{a_j}{b_j}\right), \quad \text{(B.1)}$$

where H_n *denotes the* n*-th harmonic number.*

Proof. In in view of the elementary differentiation formula

$$\frac{d}{dx}\frac{1}{(x)_k} = \frac{\psi(x) - \psi(x+k)}{(x)_k},$$

where $\psi(z) = \Gamma'(z)/\Gamma(z)$ stands for digamma function, we have:

$$x^2\left(\sum_{k=1}^{\infty} \frac{b_k}{(x)_k}\right)^2 F'(x) = \sum_{k,n=0}^{\infty} b_k a_n \frac{(\psi(x+k) - \psi(x+n))x^2}{(x)_k (x)_n}.$$

As, clearly, $(x)_k/x = (x+1)_{k-1} \to (k-1)!$ and $x(\psi(x+n) - \psi(x)) \to 1$ as $x \to$ for $n = 1, 2, \ldots$, by letting $x \to 0$ we obtain

$$\left(\sum_{k=1}^{\infty} \frac{b_k}{(k-1)!}\right)^2 F'(0+) = a_0 \sum_{k=1}^{\infty} \frac{b_k}{(k-1)!} - b_0 \sum_{k=1}^{\infty} \frac{a_k}{(k-1)!} + \sum_{k,n=1}^{\infty} b_k a_n \frac{(\psi(k) - \psi(n))}{(k-1)!(n-1)!}.$$

Finally using $\psi(n) = H_{n-1} - \gamma$, where H_n denotes the n-th harmonic number and γ is Euler-Mascheroni constant, we arrive at (B.1). □

References

[1] Baricz, Á., Vesti, J., Vuorinen, M.: On Kaluza's sign criterion for reciprocal power series. Ann. Univ. Maria Curie-Skłodowska Sect. A **65**(2), 1–16 (2011). https://doi.org/10.2478/v10062-011-0009-2

[2] Belzunce, F., Ortega, E.-M., Ruiz, J.M.: On non-monotonic aging properties from the Laplace transform, with actuarial applications. Insurance Math. Econom. **40**(1), 1–14 (2007). https://doi.org/10.1016/j.insmatheco.2006.01.010

[3] Biernacki, M., Krzyż, J.: On the monotonicity of certain functionals in the theory of analytic functions. Ann. Univ. Maria Curie-Skłodowska Sect. A **9**, 135–147 (1955)

[4] Yu, A.: Brychkov, On some properties of the Nuttall function $Q_{\mu,\nu}(a,b)$. Integr. Transf. Spec Funct. **25**(1), 34–43 (2014). https://doi.org/10.1080/10652469.2013.812172

[5] Buchstaber, V.M., Glutsyuk, A.A.: Total positivity, Grassmannian and modified Bessel functions, Functional Analysis and Geometry: Selim Grigorievich Krein Centennial, Contemp. Math., 733, AMS, 97–107. (2019) https://doi.org/10.1090/conm/733/14736

[6] Carlson, B.C., Gustafson, J.L.: Total Positivity of Mean Values and Hypergeometric Functions, SIAM J. Math. Anal. Vol. 14, No. 2, (1983). https://doi.org/10.1137/0514030

[7] Chen, W.Y.C., Louck, J.D.: The factorial Schur function, J. Math. Phys. 34, 4144. (1993) https://doi.org/10.1063/1.530032

[8] Choi, M., Smith, L.: Ordinal aggregation results via Karlin's variation diminishing property. Journal of Economic Theory **168**, 1–11 (2017). https://doi.org/10.1016/j.jet.2016.12.001

[9] Delgado, J., Orera, H., Peña, J.M.: Accurate computations with Laguerre matrices. Numer Linear Algebra Appl. **26**, e2217 (2019). https://doi.org/10.1002/nla.2217

[10] Derbazi, Z.: On Preserving or Reversing Higher-Order Unimodality and Convexity by Sign-Regular Kernels, preprint, arXiv:2502.13136, 20225. https://arxiv.org/abs/2502.13136

[11] Estrada, R., Pavlović, M.: L'Hôpital's monotone rule. Gromov's theorem, and operations that preserve the monotonicity of quotients, Publications de L'institut Mathématique Nouvelle série, Tome **101**(115), 11–24 (2017). https://doi.org/10.2298/PIM1715011E

[12] Fallat, S., Johnson, Ch.R., Sokal, A.D.: Total positivity of sums, Hadamard products and Hadamard powers: Results and counterexamples, Linear Algebra and its Applications, 520, 242–259. (2017) https://doi.org/10.1016/j.laa.2017.01.013

[13] Ferreira, R.A.C., Simon, Th.: Convolution of beta prime distribution, Trans. Amer. Math. Soc. 376, 855–890. (2023) https://doi.org/10.1090/tran/8748

[14] Gasper, G., Rahman, M.: Basic hypergeometric series, second edition, Encyclopedia of Mathematics and its Applications 96, Cambridge University Press, (2004)

[15] Grover, P., Panwar, V.S., Satyanarayana Reddy, A.: Positivity properties of some special matrices, Linear Algebra Its Appl., 596, 203–215. (2020) https://doi.org/10.1016/j.laa.2020.03.008

[16] Heikkala, V., Vamanamurthy, M.K., Vuorinen, M.: Generalized elliptic integrals. Comput. Methods Funct. Theory **9**(1), 75–109 (2009). https://doi.org/10.1007/BF03321716

[17] Ismail, M.E.H., Muldoon, M.: Higher Monotonicity Properties of q-gamma and q-psi Functions, Advances in Dynamical Systems and Applications, Volume 8, Number 2, 247–259. (2013) https://campus.mst.edu/adsa/index_files/adsa82.htm

[18] Jameson, G.J.O.: Monotonic ratios of functions. Math. Gaz. **105**(562), 129–134 (2021). https://doi.org/10.1017/mag.2021.22

[19] Kac, V., Cheung, P.: Quantum Calculus. Springer (2002). https://doi.org/10.1007/978-1-4613-0071-7

[20] Kalmykov, S.I., Karp, D.B.: Log-concavity and Turán type inequalities for the generalized hypergeometric function, Anal. Math., Volume 43, Issue 4, 567–580. (2017) https://doi.org/10.1007/s10476-017-0503-z

[21] Kalmykov, S.I., Karp, D.B.: Inequalities for some basic hypergeometric functions, Issues of Analysis (Problemy Analiza), Volume 8(26), No.1, 47–64. (2019) https://doi.org/10.15393/j3.art.2019.5210

[22] Karlin, S.: Total Positivity, Vol. I, Stanford University Press, California (1968)

[23] Karp, D., Prilepkina, E.: Completely monotonic gamma ratio and infinitely divisible H-function of Fox, Comput. Methods Funct. Theory., 16, 135–153. (2016) https://dx.doi.org/10.1007/s40315-015-0128-9

[24] Karp, D., Sitnik, S.M.: Log-convexity and log-concavity of hypergeometric-like functions, J. Math. Anal. Appl., 364, 384–394. (2010) https://doi.org/10.1016/j.jmaa.2009.10.057

[25] Koumandos, S., Pedersen, H.L.: On the asymptotic expansion of the logarithm of Barnes triple gamma function, Math. Scand. 105, no. 2, 287–306. (2009) https://doi.org/10.7146/math.scand.a-15119

[26] Mao, Zh.-X., Du, X.-Y., Tian, J.-F.: Some monotonicity rules for quotient of integrals on time scales, arXiv:2312.10252v1, December (2023). https://doi.org/10.48550/arXiv.2312.10252

[27] Mao, Zh.-X., Tian, J.-F.: Monotonicity rules for the ratio of two function series and two integral transforms, Proc. Amer. Math. Soc. 152, 2511–2527. (2024) https://doi.org/10.1090/proc/16728

[28] Mao, Zh.-X., Tian, J.-F.: Monotonicity rules for the ratio of power series, preprint (2024) arXiv:2404.18168

[29] Mao, Zh.-X., Tian, J.-F.: Monotonicity and complete monotonicity of some functions involving the modified Bessel functions of the second kind, C. R. Math., Volume 361, 217–235. (2023) https://doi.org/10.5802/crmath.399

[30] Mao, Zh.-X., Tian, J.-F.: Delta L'Hospital-, Laplace- and Variable Limit-Type Monotonicity Rules on Time Scales, Bull. Malays. Math. Sci. Soc., Volume 47, article number 1, (2024). https://doi.org/10.1007/s40840-023-01599-8

[31] Qi, F.: Decreasing properties of two ratios defined by three and four polygamma functions, C. R. Math., Volume 360, 89–101. (2022) https://doi.org/10.5802/crmath.296

[32] Pinelis, I.: L'Hôspital type rules for oscillation, with Applications, J. Inequal. Pure Appl. Math., Volume 2, Issue 3, Article 33, (2001). https://www.emis.de/journals/JIPAM/article149.html?sid=149

[33] Pólya, G., Szegö, G.: Problems and Theorems in Analysis II. Springer Science and Business Media, Mathematics (1997)

[34] Richards, D.St.P.: Totally Positive Kernels, Pólya Frequency Functions, and Generalized Hypergeometric Series, Linear Algebra Its Appl., 137/138, 467–478. (1990) https://doi.org/10.1016/0024-3795(90)90139-4

[35] Simić, S., Vuorinen, M.: Landen inequalities for zero-balanced hypergeometric functions, Abstr. Appl. Anal. 2012, Article ID 932061, 11pp. (2012) https://doi.org/10.1155/2012/932061

[36] Srivastava, H.M., Chaudry, M.A., Agarwal, R.P.: The incomplete Pochhammer symbols and their applications to hypergeometric and related functions. Integr. Transf. Spec. Funct. **23**, 659–683 (2012). https://doi.org/10.1080/10652469.2011.623350

[37] Yang, Zh.-H., Chu, Yu-M., Wang, M.-K.: Monotonicity criterion for the quotient of power series with applications, J. Math. Anal. Appl., 428, 587–604. (2015) https://doi.org/10.1016/j.jmaa.2015.03.043

[38] Yang, Z.-H., Qian, W.-M., Chu, Y.-M., Zhang, W.: Monotonicity rule for the quotient of two functions and its application, J. Inequal. Appl., article 106. (2017) https://doi.org/10.1186/s13660-017-1383-2

[39] Yang, Z.-H., Tian, J.-F.: Monotonicity rules for the ratio of two Laplace transforms with applications. J. Math. Anal. Appl. **470**(2), 821–845 (2019). https://doi.org/10.1016/j.jmaa.2018.10.034

Dmitrii Karp and Anna Vishnyakova
Department of Mathematics
Holon Institute of Technology
Holon
Israel
e-mail: dimkrp@gmail.com;
annalyticity@gmail.com

Yi Zhang
Department of Foundational Mathematics, School of Mathematics and Physics
Xi'an Jiaotong-Liverpool University
Suzhou 215123
China
e-mail: Yi.Zhang03@xjtlu.edu.cn

Received: September 10, 2024.

Accepted: April 26, 2025.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.